UNIT 3 · TOPIC 5

Testing of Special Planters

Syllabus topic: Testing of special equipment: sugarcane, cotton, rice and potato planters

Sugarcane cutter-planter

Sett length and cutting quality (**bud damage %** — blade sharpness/registration), sett rate (setts/m and overlap), placement depth, fertilizer/chemical application, missing %, field capacity vs manual planting cost.

Potato planter

Tuber spacing uniformity (miss/multiple, as for precision planters), **tuber damage/bruising %** (cup/belt handling), planting depth and ridge geometry (profilometer), plant emergence.

Rice transplanter (RNAM)

- Hills/m² and hill spacing; **seedlings per hill** distribution.
- **Missing hills %** (acceptance commonly ≤ ~5%); floating/buried/damaged hills %.
- Planting depth; mat-seedling specification compliance.
- Field capacity — typically 8-row walk-behind ~**0.15–0.2 ha/h**; riding ~**0.3–0.4 ha/h**.

Cotton planter

Metering of delinted (acid-treated)/fuzzy seed, hill spacing, seeds/hill, seed damage; check-row/hill-drop function where fitted.

Practice MCQs

Q99. In rice transplanter testing, an important acceptance parameter is:

(a) Missing hills % (≤ ~5%) (b) Seed rate kg/ha
(c) VMD (d) MWD

Q100. 'Seedlings per hill' distribution is recorded while testing a:

(a) Potato planter (b) Rice transplanter
(c) Sugarcane planter (d) Cotton drill

Q101. In sugarcane cutter-planter tests, blade condition is judged by:

(a) Bud damage % of setts (b) Paint wear
(c) Noise (d) Sett colour

Q102. Potato planter evaluation gives special attention to:

(a) Tuber bruising/damage and spacing (b) Droplet size
(c) Threshing (d) Winnowing

Q103. A riding-type rice transplanter typically achieves about:

(a) 0.05 ha/h (b) 0.3–0.4 ha/h
(c) 1.5 ha/h (d) 3 ha/h

Answers

Q99 — (a) Missing hills % (≤ ~5%)
Q100 — (b) Rice transplanter
Q101 — (a) Bud damage % of setts
Q102 — (a) Tuber bruising/damage and spacing
Q103 — (b) 0.3–0.4 ha/h

Sources: Kepner, Bainer & Barger — Principles of Farm Machinery, Ch. 2 (Field Capacities) & Ch. 17; RNAM test codes (combine, thresher, transplanter); BIS standards for power threshers and chaff cutters; Jagdishwar Sahay; NPTEL lectures.